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SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

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CA3 - Case Study Analysis Report

"MedAssist AI: An Explainable Clinical Decision Support System"

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1. Brief Problem Description

India faces a critical healthcare access crisis in rural areas. With a doctor-to-patient ratio of approximately 1:1,456 - far below the WHO-recommended 1:1,000 - millions of patients in rural communities receive no timely medical attention. Primary Health Centres (PHCs) are often understaffed, under-equipped, and geographically inaccessible to the populations they serve. ASHA (Accredited Social Health Activist) workers and community health kiosks fill a vital gap but lack decision-support tools to aid in early disease screening.

MedAssist AI addresses this critical gap through a lightweight, offline-capable Clinical Decision Support System (CDSS) designed for early disease detection. The system targets common but frequently mismanaged conditions - including diabetes, hypertension, dengue, malaria, tuberculosis, and respiratory disorders - using patient-reported symptoms as its primary input.

The enhanced version (CA3) extends the original rule-based system by integrating a Machine Learning component (Random Forest Classifier) with the existing Forward Chaining rule engine, creating a Hybrid CDSS. This dual-inference architecture improves diagnostic accuracy while retaining the interpretability and offline functionality essential for low-resource deployment environments.

1.1 Key Constraints Addressed

- No reliable internet connectivity in target rural settings
- Low-end hardware with limited processing capacity
- Non-specialist users (ASHA workers, patients) as primary operators
- Need for explainable, trustworthy outputs that clinicians can verify
- Multilingual user base requiring vernacular input support (8+ Indian languages)

2. Literature Review

The following table summarises the key research works that inform the design, architecture, and AI methodology of MedAssist AI. Each source directly maps to a component or design decision in the system.

No.	Author / Source	Title / System	Description	Relevance to MedAssist AI	Source
1	Shortliffe & Buchanan (1975)	MYCIN Expert System	Rule-based expert system for bacterial infection diagnosis using IF-THEN logic	Expert systems can encode clinical knowledge for decision support; forms the theoretical basis for MedAssist AI's rule engine	ACM Digital Library
2	Miller et al. (1982)	INTERNIST-1 / QMR	Problem-oriented diagnostic system using probabilistic associations between diseases and findings	Disease-finding probability linkage informs MedAssist AI's confidence scoring and differential ranking	NEJM / PubMed
3	Breiman (2001)	Random Forests	Ensemble learning method combining multiple decision trees for classification with improved accuracy	Directly implemented in MedAssist AI as the ML classifier for symptom-to-disease mapping using 132 symptom features	Machine Learning Journal
4	Musen et al. (2014)	Clinical Decision Support Systems	Comprehensive review of CDSS architecture, challenges, and adoption in low-resource settings	Validates the modular architecture and knowledge base design used in MedAssist AI; highlights need for offline capability	JAMIA
5	WHO ICD-11 (2022)	ICD-11 Clinical Guidelines	International Classification of Diseases providing standardised disease codes, diagnostic criteria, and symptom mappings	Core knowledge source for MedAssist AI's rule base; ensures medical validity and alignment with global standards	who.int
6	ICMR Protocol (2023)	India-Specific Clinical Protocols	ICMR guidelines for common diseases in India including TB, dengue, malaria, hypertension, and diabetes	Provides India-specific disease rules for rural relevance; supplements ICD-11 for local disease prevalence	icmr.gov.in
7	Buch et al. (2018)	Artificial Intelligence in Medicine	Systematic review of ML techniques applied in clinical settings including classification, diagnosis, and explainability	Supports choice of Random Forest over deep learning for explainability and offline deployment constraints	Nature Medicine
8	Adadi & Berrada (2018)	Explainable AI (XAI)	Survey of interpretability techniques in AI systems for healthcare, including feature importance and rule extraction	Supports MedAssist AI's explainable output design - confidence scores, key symptoms, and reasoning transparency	IEEE Access

3. AI-Based Solution Approach

MedAssist AI employs a hybrid dual-inference pipeline combining rule-based reasoning and machine learning prediction, connected through a confidence fusion layer. The system follows six structured steps from input to explainable output.

Step	Stage	Description
Step 1	**Data Collection & Preprocessing**	The Kaggle Disease-Symptom Description Dataset (132 symptoms × 41 diseases) is used. Symptoms are encoded into a binary vector where 1 = present and 0 = absent. Patient demographic data (age, gender, pre-existing conditions) is also collected via the Symptom Collector interface.
Step 2	**Knowledge Representation**	Medical expertise is encoded using two complementary representations: (i) Semantic Networks map relationships between concepts (IS-A, HAS-A, PRODUCES, USES, CAPTURES), and (ii) Frame Representation stores attribute-level details for each entity (Patient Profile, Knowledge Base, Inference Engine, Diagnosis Output) - providing both structural and attributional knowledge.
Step 3	**Rule-Based Inference (Forward Chaining)**	The Rule Engine applies Forward Chaining over IF-THEN rules derived from WHO ICD-11 and ICMR protocols. Starting from known patient facts, the engine matches symptom combinations to disease conditions and generates initial diagnoses with confidence weights. Example rule: IF [fever AND joint_pain AND rash] THEN [Dengue - confidence: 0.85].
Step 4	**Machine Learning Prediction**	A Random Forest Classifier (scikit-learn) processes the same binary symptom vector in parallel. The model outputs a probability distribution across 41 diseases. Random Forest is chosen for its feature importance (supporting explainability), robustness to noise, and suitability for offline deployment on low-end hardware.
Step 5	**Confidence Fusion Layer**	Outputs from both tracks are merged via a weighted scoring formula: Final Score = (w_1 × ML Score) + (w_2 × Rule Score). This hybrid fusion leverages the interpretability of rules and the pattern-recognition power of ML. Diseases are ranked by final confidence score; the top 3 are presented to the user.
Step 6	**Explainable Output Generation**	The Diagnosis Output module presents: (i) Top 3 probable diseases with confidence percentages, (ii) Key symptoms contributing to each diagnosis, (iii) Urgency/triage level (Low / Moderate / High), and (iv) Recommended next steps. All output includes a disclaimer stating it is a preliminary screening tool.

3.1 System Architecture Flow

The high-level architecture follows a linear modular pipeline:

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User Interface → Symptom Collector → Patient Profile → Inference
Engine → Confidence Aggregator → Diagnosis Output
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The Inference Engine itself runs two parallel tracks: (i) Rule-Based Forward Chaining and (ii) Random Forest ML Prediction. Their outputs converge at the Confidence Aggregator, which applies weighted fusion before generating the final ranked diagnosis list.

3.2 Tech Stack Summary

- Frontend: Streamlit (Python) - interactive web-based UI for symptom input and diagnosis display
- Backend: FastAPI (Python) - handles API requests, ML model, and rule engine
- ML Model: scikit-learn Random Forest Classifier - trained on 132-symptom binary feature vectors
- Rule Engine: Custom IF-THEN Forward Chaining engine (Python)
- Database: SQLite (offline local storage) - enables full offline functionality
- Dataset: Kaggle Disease-Symptom Description Dataset (132 symptoms × 41 diseases)

4. Justification & Reasoning

4.1 Why a Hybrid AI Approach?

A purely rule-based system, while interpretable, is limited by the completeness of its hand-crafted rules and cannot generalise well to atypical symptom presentations. A purely ML-based system, while accurate, operates as a black box - inappropriate in a clinical context where trust and explainability are critical. The hybrid approach balances both: rules provide medically grounded, interpretable baseline diagnoses, while the Random Forest adds statistical pattern-recognition over the full 132-symptom feature space, capturing co-occurrences that individual rules may miss.

4.2 Why Random Forest?

Random Forest is selected over alternatives (e.g., neural networks, SVM) for the following clinically and technically justified reasons:

- Handles high-dimensional binary data (132 symptom features) efficiently without dimensionality reduction
- Provides built-in feature importance scores - enabling explainability of why a diagnosis was suggested
- Robust to noise and class imbalance common in medical datasets
- Works well with limited preprocessing and no requirement for data normalisation
- Suitable for offline deployment on low-end devices without GPU acceleration

4.3 Why Forward Chaining?

Forward Chaining (data-driven inference) is the appropriate reasoning strategy because the system begins with known patient facts (symptoms) and derives conclusions (diagnoses). This contrasts with backward chaining (hypothesis-driven), which would require a target disease hypothesis upfront - not feasible for general-purpose screening. Forward chaining allows the system to progressively match all applicable rules and surface multiple candidate diagnoses, ranked by confidence.

4.4 Why Semantic Networks and Frame Representation?

Semantic Networks are used to model the structural relationships between system components (IS-A, HAS-A, PRODUCES, USES). This enables inheritance - MedAssist AI inherits all CDSS properties from its parent concept while adding India-specific extensions. Frame Representation complements this by encoding rich attribute-level descriptions, including storage mechanisms, sync behaviour, privacy constraints, and conflict resolution strategies. Together, these two knowledge representation techniques enable structured, medically grounded reasoning over incomplete patient data.

4.5 Why Offline-First Architecture?

Rural India's connectivity challenges make internet-dependent systems impractical for consistent deployment. The offline-first design using SQLite local storage, service workers, and a locally hosted FastAPI backend ensures full system functionality without network access. This is a deliberate design choice aligned with the deployment context - primary health centres, ASHA worker tablets, and community health kiosks - rather than a technological compromise.

5. Individual Contributions

The project was collaboratively developed with clearly defined responsibilities for each team member. All members contributed equally to the final report, presentation, and review.

Name (PRN)	Contribution Details
Parth Damle (23070122161)	Worked on backend development, knowledge representation, dataset selection and documentation.
Pratik Lakra (23070122166)	Worked on frontend development, rule-based inference design and dataset selection.
Rajat Singh (23070122173)	Worked on backend development, inference engine integration, and dataset selection.

5.1 Collective Achievements

As a team, we successfully designed a complete AI-based clinical decision support system grounded in formal knowledge representation (Unit 3), hybrid AI techniques including rule-based and ML inference (Unit 4), and a detailed case analysis with justification mapped to India's rural healthcare context (Unit 5). The system architecture demonstrates practical application of course concepts in a socially relevant domain.

5. References

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Research Paper Links:

- Shortliffe & Buchanan (1975) - MYCIN: https://dl.acm.org/doi/10.1145/1499586.1499643
- Breiman (2001) - Random Forests: https://link.springer.com/article/10.1023/A:1010933404324
- Musen et al. (2014) - CDSS Review: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3994985/
- Buch et al. (2018) - AI in Medicine: https://www.nature.com/articles/s41591-018-0107-6
- WHO ICD-11 (2022): https://icd.who.int/en
- ICMR Guidelines (2023): https://www.icmr.gov.in/guidelines

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